Channel-to-Channel Crosstalk in a 32-Channel Dry-Electrode ADS1299 Cap

A single-driven-input measurement, and why it sets only an upper bound

EEG_MI project — hardware metrology

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Summary

A 7 Hz sine from an external generator was driven into exactly one channel input of the 32-channel cap, off the head, with GND and REF tied to the generator's ground and every other channel input left open. The question was how much of that signal leaks into the other channels.

No other channel shows a detectable tone. The apparent 0.6 µV to 1.1 µV of 7 Hz on the channels that were not saturated is statistically indistinguishable from the amplitude the same estimator returns at *arbitrary* nearby frequencies ($p = 0.20$–0.51 against a 75-point null distribution), and the broadband correlation between the driven channel and every other channel is $r \leq +0.002$. What the measurement actually delivers is therefore an *upper bound*:

Crosstalk into any measurable channel	≤ -24.1 dB (bound, not an estimate)
Driven channel amplitude	24.542 µV peak (17.354 µV rms)
Driven channel noise, 20 Hz to 45 Hz	0.092 µV rms
Driven channel THD ($2f$–$6f$)	0.22 %
Driven channel mains (50 Hz)	0.0045 µV
Tone frequency	7.0226 Hz (+3235 ppm vs nominal)

The bound is weak, and the reason matters: it is set by the noise of the *unconnected* inputs, not by the amplifier. The one properly terminated input has a noise-only detection ceiling of 0.066 µV, so repeating this exact record with every unused input tied to generator ground would bound crosstalk at about −51 dB — 27 dB better, at no cost but a few jumper wires. Section 7 gives the protocol.

Two estimator faults were found and corrected in the process, both of which had produced confidently wrong answers (Section 3). They are documented here because either one silently converts a null result into a dramatic one.

1 Setup

Hardware

Amplifier	TI ADS1299, 24 bit, gain 24, 250 Hz
Scaling	µV = counts × 0.02235; full scale ±187 500 µV
Cap	32-channel dry electrode, 10–20 montage, VHDCI-68 interface
Transport	WiFi UDP, 105-byte frames, 1-byte sequence counter
Source	external signal generator, 7 Hz sine, amplitude not set by the operator

Wiring — and the part that determines what can be concluded

generator ground	→ board GND *and* REF
generator output	→ **one** channel input
all other channel inputs	**open**

An ADS1299 input with nothing attached has no DC path to set its operating point, so it drifts to a rail. That is exactly what happened: 25 of the 32 inputs sit pinned at +187 500 µV for 100 % of the record. A channel at the rail is not a quiet channel — it is a channel carrying no information at all, and it cannot show crosstalk even if crosstalk exists. Six further inputs had drifted to 98 mV to 180 mV without saturating; those are high-impedance nodes, i.e. antennas.

The driven channel was not declared by the operator. It is identified from the data as **FZ** (VHDCI pin 31): it is the only input at 0 V DC, it carries the largest tone by a factor of 20, its noise floor is 100× lower than any other channel, and its 50 Hz content is 1218× lower. All four are the signature of an input connected to a low-impedance source.

Table 1: Input state. Only one of 32 inputs was in a condition where a crosstalk measurement is meaningful.

State	Criterion	n	Channels
railed	> 50 % of samples at $\geq 0.97\times$ full scale	25	FP2, AF4, F3, F4, F7, F8, FC1, FC2, FC5, FC6, C3, C4, T7, T8, CP1, CP2, CP5, CP6, P3, P4, P7, PO3, O2, CZ, P
floating	$\lvert\overline{V}\rvert > 1$ mV, not railed	6	FP1, AF3, P8, PO4, O1, OZ
terminated	$\lvert\overline{V}\rvert \leq 1$ mV	1	FZ

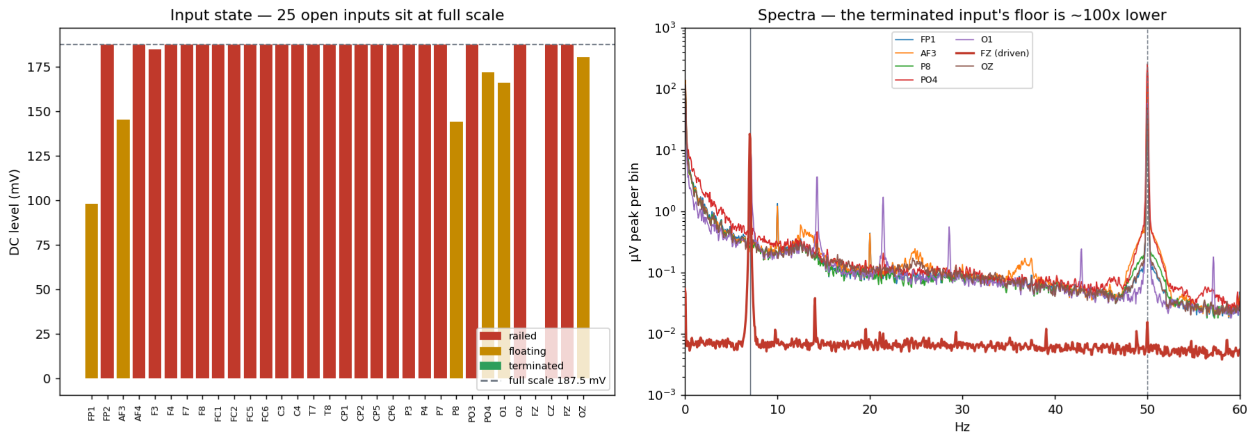

Figure 1: Left: DC level per input. Twenty-five inputs sit exactly at full scale. Right: amplitude spectra of the seven channels that are not saturated. The terminated channel (red) has a noise floor roughly two decades below the floating ones and essentially no 50 Hz; the floating channels carry 3.5 µV to 26.5 µV of mains and a large low-frequency drift skirt.

2 The driven channel

Before asking about leakage, it is worth recording how the one properly connected channel behaves, because it is the only clean characterisation of the amplifier in this dataset.

Two remarks. First, 0.092 µV rms over 25 Hz of bandwidth is consistent with the ADS1299 datasheet's input-referred noise at gain 24, so the amplifier is behaving as specified when its input is properly terminated. Second, the +3235 ppm frequency offset is the *combined* error of the generator's timebase and the cap's sample clock and cannot be attributed between them from this record alone. It is far too large for a crystal (≈20 ppm), so most of it belongs to one of the two oscillators being nominal-rather-than- calibrated; a generator set to "7 Hz" by a DIP switch is the more likely

Table 2: Driven channel (FZ), 179 s record.

Amplitude	24.542 µV peak	17.354 µV rms
Amplitude stability	24.05 ± 0.03 µV	across 35 consecutive 5 s windows
Frequency	7.0226 Hz	+3235 ppm vs the nominal 7 Hz
THD ($2f$–$6f$)	0.22 %	
50 Hz	0.0045 µV	
Noise, 20 Hz to 45 Hz	0.092 µV rms	
Noise-only ceiling at f_0	0.066 µV	95th pct of the null distribution

candidate. Either way it is small enough not to matter for EEG band power, and large enough to destroy a fixed-frequency fit (Section 3).

3 Two estimator traps

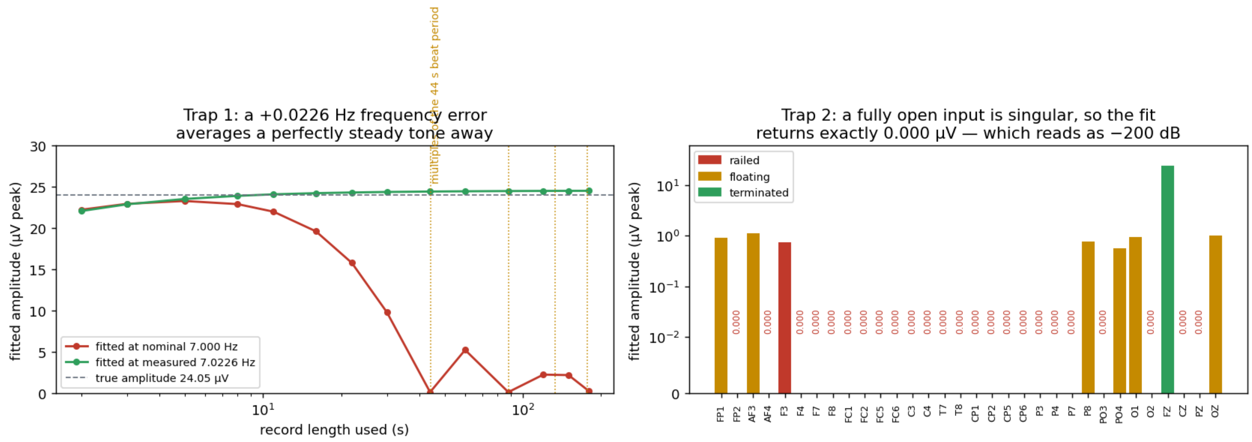

Figure 2: Left: the same steady 24.05 µV tone, fitted at the nominal 7.000 Hz (red) and at the measured 7.0226 Hz (green), as a function of how much of the record is used. The red curve collapses at multiples of the 44 s beat period. Right: fitted amplitude per channel. The 24 fully saturated inputs return exactly 0.000 µV because the least-squares system is singular — which a decibel conversion turns into −200 dB, i.e. "perfect isolation".

Trap 1: fitting the nominal frequency. The tone is at 7.0226 Hz, not 7.000 Hz. A least-squares projection onto a fixed 7.000 Hz sinusoid accumulates $2\pi \times 0.0226\,t$ of phase error, so over a record spanning several 44 s beat periods the projection averages the tone away. Measured on this record, the same signal reads:

Record length	11 s	59 s	179 s	5 s windows
fitted at 7.000 Hz	21.84 µV	5.17 µV	0.31 µV	24.05 µV
fitted at 7.0226 Hz	24.5 µV	24.5 µV	24.54 µV	24.05 µV

Read down the first row and the source appears to be dying exponentially across three successive recordings. It is not: it is flat to ±0.1 % throughout. The fix is to estimate the frequency once on the driven channel and use that same value for every channel, so source and victims are measured on the same basis.

Trap 2: railed channels read as zero. For a channel pinned at a constant value the least-squares design matrix is singular and the fit returns exactly 0.000 µV. Converted to decibels that becomes −200, which is indistinguishable in a results table from "no crosstalk". In the first run of this measurement 24 channels landed in that state, and the worst-case crosstalk figure was then computed across the six *floating* channels only — producing a headline "−4.8 dB worst-case crosstalk" that was an artifact end to end. Open inputs must be reported as unmeasurable, not as zero.

4 Is the tone detectable on any other channel?

An amplitude fit always returns a positive number. On a drifting channel with a 1 µV-scale noise skirt it returns roughly 1 µV whatever frequency is requested, so an absolute value alone establishes nothing. The test used here builds the null from the channel's own noise: fit the amplitude at 75 frequencies spread over 4 Hz to 10 Hz, excluding f_0 and its first three harmonics, and ask what fraction of those null frequencies return an amplitude at least as large as the one at f_0.

Table 3: Detection test, 179 s record, 75 null frequencies. **Bound** is the null 95th percentile expressed in dB relative to the 24.542 µV source: the smallest leakage this channel could have revealed.

Ch	State	$A(f_0)$ µV	null med µV	null p95 µV	z	p	Bound dB	Verdict
FZ	terminated	24.542	0.027	0.066	1549.3	0.000	−51.3	**tone present**
O1	floating	0.955	0.702	1.015	1.1	0.200	−27.7	not distinguishable
FP1	floating	0.899	0.773	1.257	0.4	0.427	−25.8	not distinguishable
OZ	floating	0.995	0.855	1.382	0.5	0.427	−25.0	not distinguishable
AF3	floating	1.131	0.970	1.522	0.5	0.440	−24.1	not distinguishable
P8	floating	0.781	0.696	1.157	0.3	0.453	−26.5	not distinguishable
PO4	floating	0.576	0.583	0.927	0.2	0.507	−28.5	not distinguishable
25 railed channels: unmeasurable (no information at any frequency)								

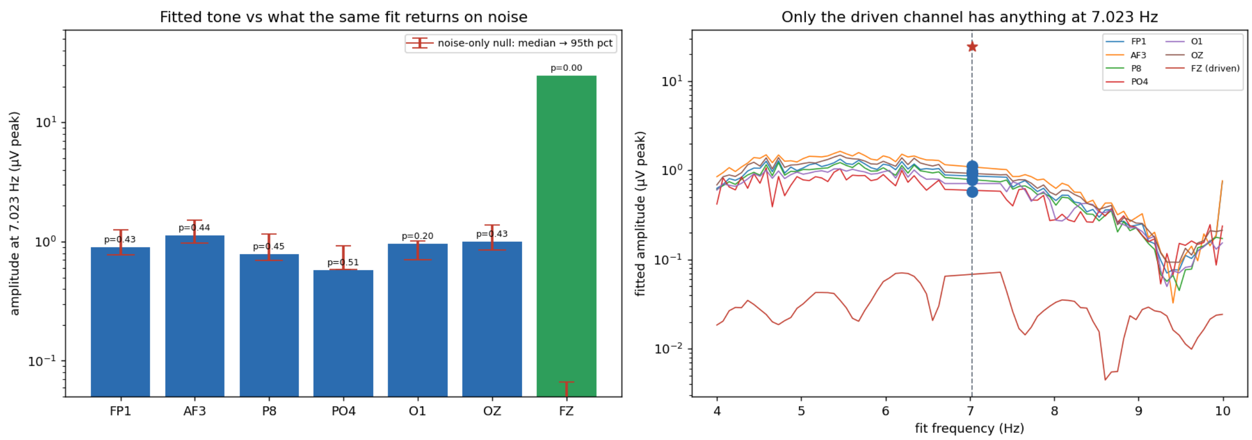

Figure 3: Left: fitted amplitude at f_0 per channel (bars) against the noise-only null median and 95th percentile (red). Every floating channel's "signal" sits inside its own noise distribution. Right: the same fit swept across frequency. The floating channels trace a smooth ∼1 µV curve with **no feature at** f_0; the driven channel sits at 0.02 µV to 0.07 µV everywhere except the single point at 7.023 Hz. The right panel is the whole result in one image.

5 Controls

Three quantities looked, at first, like evidence of a shared coupling path. Each dissolves under its own control.

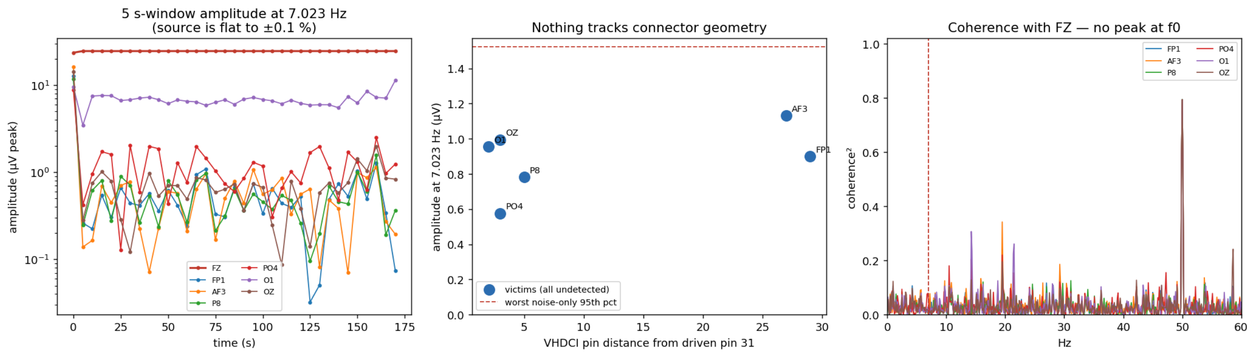

Figure 4: Left: 5 s-window amplitude at f_0; the source is flat, the floating channels wander around 1 µV with no relationship to it. Centre: amplitude against VHDCI pin distance from the driven pin. Right: magnitude-squared coherence with the driven channel — no peak at f_0.

Phase clustering. The six floating channels' phases at f_0 cluster within 9.3° (circular sd) of each other, which looks like one shared path. But the same statistic computed at the 75 null frequencies has a median of 4.4° — *tighter* — and $p = 0.787$. The floating inputs are phase-locked to one another at essentially every frequency because they share the same ambient field (median pairwise broadband $r = 0.44$). A tight cluster at f_0 is therefore expected under the null and carries no information.

Correlation with the source. The decisive control. Broadband Pearson correlation between the driven channel and each floating channel: +0.0011, +0.0011, +0.0015, +0.0018, +0.0021, +0.0022. The driven channel is uncorrelated with every other channel in the cap.

Harmonics. The floating channels return 25 % to 45 % "THD", against 0.22 % on the source. Read naively this says the coupling path generates harmonics and is therefore nonlinear. It says nothing of the kind: those are fits at 14 and 21 Hz to the same broadband noise that produced the fundamental, and a harmonic ratio is meaningless while the fundamental itself is undetected.

Connector geometry. Amplitude versus VHDCI pin distance from the driven pin 31 gives $r = +0.47$ over $\Delta\text{pin} = 2\text{–}29$ — weakly *increasing* with distance, the opposite of what ribbon lead-to-lead capacitance would produce. FP1 at $\Delta\text{pin} = 29$ reads more than PO4 at $\Delta\text{pin} = 3$. Consistent with there being no geometric coupling to find.

6 What this does and does not establish

Established. Crosstalk from the driven channel into any measurable channel is below −24.1 dB (worst channel AF3). The amplifier, when its input is terminated, has a 0.092 µV rms noise floor, 0.22 % THD and negligible mains pickup. No signal path was found between the driven channel and any other.

Not established. Whether crosstalk is −30 dB or −90 dB. The bound is set by the *victims'* noise, and 31 of them were unconnected. A −24 dB bound is far too loose to be a hardware specification: a 50 µV alpha rhythm on one channel would be permitted to put 3 µV onto its neighbour, which would be a serious defect if real. Nothing here says it is real; the measurement simply could not have seen it.

Also not established. Any statement about coupling *mechanism*. With no detectable leakage there is nothing whose mechanism could be identified, and the phase, harmonic and geometry patterns above are all noise.

7 Protocol for a real crosstalk specification

1. **Terminate every unused input to generator ground.** This is the single change that matters. It takes the victim noise floor from ∼1.0 µV to ∼0.07 µV and the bound from −24 to about −51 dB, on the same 179 s record and the same source amplitude.
2. **Use the largest calibrator output.** At 2000 µV instead of 24.5 µV the bound improves by a further 38 dB, to roughly −89 dB — below any level that could matter for EEG.
3. **Measure the tone frequency; never assume the nominal.** One `refine()` call on the driven channel, then fit every channel at that frequency.
4. **Report open inputs as unmeasurable.** Never let a singular fit become −200 dB.
5. **Move the driven channel and repeat.** Drive a pin at one end of the connector, then the middle, then the other end. If leakage ever becomes detectable, the dependence on Δpin separates ribbon lead-to-lead capacitance (falls off with distance) from a shared REF/BIAS return (does not).
6. **Run the generator on battery** and keep leads short and twisted; a charger ties the generator to mains earth and creates a ground loop through the amplifier.

8 How this compares to the silicon, and to normal practice

Two questions were checked against external sources: is a railing open input normal, and is the hardware acceptable on crosstalk?

The chip already has a crosstalk specification

TI's ADS1299 datasheet (SBAS499C, Table 7.5 Electrical Characteristics) publishes the numbers below. They are silicon-level typicals at gain 12, 250 SPS, and the crosstalk figure is the one that matters here.

Table 4: ADS1299 datasheet vs what this measurement could see.

Parameter	Datasheet	Test condition	This measurement
Crosstalk	**−110 dB** typ	$f_{\mathrm{IN}} = 50/60$ Hz	bound only, −24.1 dB
CMRR	−120 dB typ (−110 min)	$f_{\mathrm{CM}} = 50/60$ Hz	not measured
THD	−99 dB	$f_{\mathrm{IN}} = 10$ Hz	−53 dB (generator-limited)
Input-referred noise	1 µVPP typ, 1.35 max	0.01–70 Hz, gain 24	≈ 1.0 µVPP equivalent
Input bias current	±300 pA	$T_A = 25\,^\circ$C	—
DC input impedance	1000 MΩ	no lead-off	—
Input capacitance	20 pF		—

The comparison that matters: **the chip's own crosstalk specification is −110 dB, and this measurement could only bound crosstalk at −24 dB** — 86 dB short of being able to test it. That is not a criticism of the hardware; it is a statement about what a measurement with 31 unconnected inputs can resolve. Board layout and the VHDCI cable can of course add coupling on top of the silicon, and that is exactly what a proper terminated-input measurement would look for.

The noise comparison is the one place where this record does reach the datasheet. Measured 0.092 µV rms over 20–45 Hz (25 Hz of bandwidth); scaling a flat spectrum to the datasheet's 0.01–70 Hz band gives 0.154 µV rms, and the conventional 6.6σ peak-to-peak conversion gives 1.02 µVPP against a datasheet typical of 1 µVPP and a maximum of 1.35. The extrapolation is mildly optimistic — real amplifier noise has a $1/f$ region below 10 Hz that a 20–45 Hz measurement does not see, so the true broadband figure will sit somewhat above 1.02 µVPP. Even so, the terminated channel is operating within the datasheet's typical-to-maximum window. The analog front end is doing its job.

Railing on an open input is expected, and is documented as such

The ADS1299 datasheet's pin table carries the instruction directly, as footnote (2) on every `INxP`/`INxN` pin: *"Connect unused analog inputs directly to AVDD."* Leaving them open is not a supported configuration.

The mechanism is in the same tables. An open input has no DC return, so the ±300 pA input bias current has nowhere to go but into the 20 pF input capacitance:

$$\frac{dV}{dt} = \frac{I_{\text{bias}}}{C_{\text{in}}} = \frac{300\,\text{pA}}{20\,\text{pF}} \approx 15\,\text{V/s},$$

which drives the node into the supply rail within milliseconds. Twenty-five channels sitting at exactly full scale is therefore not a fault — it is the textbook response of this part to the wiring used.

This also matches what the wider community reports for the same silicon. The OpenBCI Cyton (also ADS1299-based) generates a steady stream of "RAILED" reports traced to floating or poorly-referenced inputs, and the standard advice given there is that all EEG amplifiers behave this way with floating pins because the front ends are extremely high impedance, and that the diagnostic is to short channel, reference and bias together — which should read close to zero microvolts. That is precisely the state our FZ channel was in, and it read 0.092 µV rms.

Crosstalk is not a commonly published EEG specification

Worth knowing when deciding what to compare against: commercial research EEG amplifiers generally publish CMRR rather than channel-to-channel crosstalk. Brain Products' actiCHamp quotes 100 dB common-mode rejection per IEC 60601-2-26, and the IEC standard for electroencephalographs is framed around amplitude accuracy, input dynamic range, input noise and frequency response. So the ADS1299's −110 dB is a more specific number than most amplifier vendors give, and there is no widely used pass/fail industry threshold for EEG crosstalk to check this cap against. The practical threshold is the application one: leakage should be far below the smallest signal of interest, and for MI work that means a 50 µV rhythm on one channel must not put a meaningful fraction of a microvolt onto its neighbour — roughly −40 dB or better as a working requirement.

Bottom line on hardware quality

Nothing in this measurement indicates a crosstalk problem, and one channel's worth of evidence says the analog front end meets its datasheet noise specification. But the measurement cannot certify the cap: a −24 dB bound is looser than the −40 dB working requirement, so "no problem found" is not yet "verified good". The terminated-input protocol in Section 7 closes that gap and costs one evening.

Sources

- Texas Instruments, *ADS1299-x Low-Noise, 4-, 6-, 8-Channel, 24-Bit ADCs for Biopotential Measurements*, SBAS499C, rev. January 2017 — Section 7.5 Electrical Characteristics (crosstalk, CMRR, THD, input-referred noise, input bias current, input capacitance, DC input impedance) and Section 6 Pin Functions ("Connect unused analog inputs directly to AVDD"). https://www.ti.com/lit/ds/symlink/ads1299.pdf
- OpenBCI community forum, railed-channel threads on floating and poorly-referenced inputs in the ADS1299-based Cyton board. https://openbci.com/forum/index.php?p=/discussion/3056/electrode-cap-all-channels-are-railed-resolved

- Brain Products GmbH, actiCHamp series specifications (CMR 100 dB per IEC 60601-2-26). https://www.brainproducts.com/solutions/actichamp/
- IEC 60601-2-26:2012, *Medical electrical equipment — Part 2-26: Particular requirements for the basic safety and essential performance of electroencephalographs*. https://webstore.iec.ch/en/publication/2637

Data and reproduction

`recordings/calibrator/cal_7hz_180s.npz`	179 s, the record analysed here
`recordings/calibrator/cal_7hz_xt7hz_60s.npz`	59 s
`recordings/calibrator/cal_7hz_probe11s.npz`	11 s
`results/crosstalk7.json`	every number in this report

```
python src/acquisition/crosstalk_analysis.py \
    recordings/calibrator/cal_7hz_180s.npz --freq 7 --json results/crosstalk7.json
```

All recordings are raw µV, unfiltered, pre-CAR, all 32 channels retained.